USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1) Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2) For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3) Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1)b)
4) Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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Country: _____ Zip/Postal Code: _____

Phone: _____ FAX: _____

Name: _____ Email: _____

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: VPD test update – TEST.PD.VDM.CBL1
Applied to: USB PD CTS Q3, 2026

Brief description of the functional changes proposed:
VPD does not support alternate mode operation and the VIF tool also greys out the Modal_Operation_Supported field setting. The test needs to be revised to stop the test sequence for VPD product type after Discover SVID has been responded with NAK.

Benefits as a result of the proposed changes:
VPDs with no Alt mode support correctly tested.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
Tester to update the corresponding tests per the ECR updates.

An analysis of the Vendor Info File (VIF) implications:
None – propose updating “Modal_Operation_Supported” to “Alternate_Modes_Supported” per the updated definition in the approved ECR, USBPD ECR Clean up Alternate Mode alias language.pdf https://groups.usb.org/wg/powerdelivery/document/18474

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested
Update Test Conditions table in the sections below.

To Text:

TEST.PD.VDM.CBL.1 Discovery Process and Enter Mode

Description: The Tester verifies that the UUT responds correctly to the Discovery Process and the *Enter Mode* Commands.

Test Conditions:

	Consumer Only	Provider Only	DRP, C/P, P/C	Free Cable	VPD, Captive Cable, Si-Only eMarker
Rev2VcMinEnd1				✓	✓
Rev3VcMinEnd1				✓	✓
Rev2VcMaxEnd1				✓	✓
Rev3VcMaxEnd1				✓	✓
Rev2VcMinEnd2				✓	
Rev3VcMinEnd2				✓	
Rev2VcMaxEnd2				✓	
Rev3VcMaxEnd2				✓	

Test Procedures:

1. There are 4 possible bring-up procedures:
    a. The UUT has VIF field VIF_Product_Type set to Cable. The Tester runs bring-up procedure with the UUT as a Cable Plug COMMON.PROC.BU.3
    b. The UUT has VIF field VIF_Product_Type set to Port Product and PD_Port_Type set to eMarker. The Tester runs bring-up procedure with the UUT as a VPD COMMON.PROC.BU.10
    c. The UUT has VIF field PD_Port_Type set to DRP, Consumer/Provider or Consumer only, and VIF parameter Captive_Cable is set to YES, and VIF parameter eMarked Captive Cableis set to YES*. The Tester runs bring-up procedure with the UUT as a Sink UUT for Captive Cable Test COMMON.PROC.BU.6.
    d. The UUT has VIF field PD_Port_Type set to Provider/Consumer or Provider only, and VIF parameter Captive_Cable is set to YES, and VIF parameter eMarked Captive Cable is set to YES*. The Tester runs bring-up procedure with the UUT as a Source UUT for Captive Cable Test COMMON.PROC.BU.11.
2. The Tester sends a *Discover Identity* Command to the UUT, using SOP'.
3. The Tester checks the response message: [TEST.PD.VDM.CBL.1#1]
    a. The check fails if the UUT responds anything but a *Discover Identity* Response message.
    b. If the UUT responds with "Responder BUSY" message, the Tester resends *Discover Identity* Command for *nBusyCount* times every *tVDMBusy* min. Once the Tester has repeated *Discover Identity* for *nBusyCount* times, the check fails.
    c. If the UUT responds with a "Responder NAK", the check fails.
    d. Check the "Responder ACK" message:
        i. In PD2 mode
            The check fails if the Number of Data Objects is not 5.
        ii. In PD3 mode
            If Product Type (ID Header B29…27) = Passive Cable, the check fails if the Number of Data Objects is not 5.

If Product Type (ID Header B29…27) = Active Cable, the check fails if the Number of Data Objects is not 6.

If Product Type (ID Header B29…27) = VPD, the check fails if the Number of Data Objects is not 5.

iii. ID Header VDO:
   1. The check fails if B31 does not match VIF field Data_Capable_as_USB_Host
   2. The check fails if B30 does not match VIF field Data_Capable_as_USB_Device
   3. The check fails if B29…27 does not match VIF field Product_Type
   4. The check fails if B26 does not match VIF field Modal_Operation_Supported
   5. In PD2, the check fails if B25…16 is not set to zero.
   6. In PD3, the check fails if B25…23 is not set to zero.
   7. In PD3, the check fails if B22…21 does not match VIF field ID_Header_Connector_Type
   8. In PD3, the check fails if B20…16 is not set to zero.
   9. The check fails if B15…0 does not match VIF field USB_VID

iv. The check fails if Cert Stat VDO does not match VIF field XID.

v. The check fails if Product VDO does not match VIF fields PID and bcdDevice

vi. Passive Cable VDO consistency with respective VIF fields:
   1. B31…28: Cable_HW_Vers
   2. B27…24: Cable_FW_Vers
   3. B23…21:
      a. In PD2 mode: Reserved 000b
      b. In PD3 mode: VDO Version is 000b.
   4. B20: Reserved 0b
   5. B19…18:
      a. In PD2 mode: Type_C_To_Type_A_B_C
      b. In PD3 mode: Type_C_To_Type_C_Capt_Vdm_V2
   6. B17:
      a. In PD2 mode: (Reserved) 0b
      b. In PD3 mode: EPR Mode Capable
   7. B16…13 Cable_Latency
   8. B12…11 Cable_Termination_Type
   9. In PD2 mode: B10…7: Any Value Permitted
   10. In PD3 mode
      a. B10…9: Max_VBUS_Voltage
      b. B10…9: If EPR Mode Capable is set to 1b, check that Max_VBUS_Voltage is set to 11b(50V)
      c. B8…7: Reserved 00b
   11. B6…5
      a. In PD2: VBUS_through_cable is YES: Cable_VBUS_Current
      b. In PD2: VBUS_through_cable is NO: (Reserved) 00b
      c. In PD3: Cable_VBUS_Current
      d. In PD3: If EPR Mode Capable is set to 1b, check that Cable_VBUS_Current is set to 10b(5A)
   12. B4
      a. In PD2: VBUS_through_cable
      b. In PD3: (Reserved) 0b
   13. B3 (Reserved) 0b
   14. B2…0 Cable_Superspeed_Support (in PD2 only)
   15. B2…0 Cable_USB_Highest_Speed (in PD3 only)

vii. Active Cable VDO consistency (in PD2) or Active Cable VDO 1 consistency (in PD3) with respective VIF fields
   1. B31…28: Cable_HW_Vers
   2. B27…24: Cable_FW_Vers
   3. B23…21:

a. In PD2: Reserved 000b
b. In PD3: VDO Version is 011b.

4. B20: Reserved 0b
5. B19…18:
   a. In PD2: Type_C_To_Type_A_B_C
   b. In PD3: Type_C_To_Type_C_Capt_Vdm_V2
6. B17:
   a. In PD2 mode: (Reserved) 0b
   b. In PD3 mode: EPR Mode Capable
7. B16…13 Cable_Latency
8. B12…11 Cable_Termination_Type
9. In PD2: B10…7: Any Value Permitted
10. In PD3:
    a. B10…9: Max_VBUS_Voltage
    b. B10..9: If EPR Mode Capable is set to 1b, check that Max_VBUS_Voltage is set to 11b(50V)
    c. B8: SBU_Supported
    d. B7: SBU Active
11. B6…5
    a. If VBUS_Through_ Cable is NO, Reserved 00b.
    b. If VBUS_Through_ Cable is YES, Cable_VBUS_Current
    c. In PD3: If EPR Mode Capable is set to 1b, check that Cable_VBUS_Current is seto to 10b(5A)
12. B4 VBUS_through_cable
13. B3 Cable_SOP"_controller
14. B2…0 Cable_Superspeed_Support (in PD2 only)
15. B2…0 Cable_USB_Highest_Speed (in PD3 only)

viii. Active Cable VDO 2 consistency in PD3

1. B31…24 Act_Cab_Max_Op_Temp
2. B23…16 Act_Cab_ShutDn_Temp
3. B15 Reserved 0b.
4. B14…12 Act_Cab_U3_CLd_Power
5. B11 Act_Cab_U3_U0_Trans_Mode
6. B10 Act_Cab_Phys_Connect
7. B9 Act_Cab_Active_Elem
8. B8 Act_Cab_USB4_Supp
9. B7…6 Act_Cab_USB_2_0_Hub_Hops
10. B5 Act_Cab_USB_2_0_Supp
11. B4 Act_Cab_USB_3_2_Supp
12. B3 Act_Cab_USB_Lanes
13. B2 Act_Cab_Opt_Isol
14. B1 USB4 Asymetric Mode Supported.
15. B0 Act_Cab_USB_Gen

ix. VPD consistency in PD3

1. B31…28 VPD_HW_Vers
2. B27…24 VPD_FW_Vers
3. B23…21 VDO Version is 000b.
4. B20…17 Reserved 0000b.
5. B16…15 VPD_Max_VBUS_Voltage
6. B14
   a. VPD_Charge_Through_Support = 1: VPD_Charge_Through_Current
   b. VPD_Charge_Through_Support = 0: Reserved 0b
7. B13 Reserved 0b.
8. B12…7
   a. VPD_Charge_Through_Support = 1: VPD_VBUS_Impedance

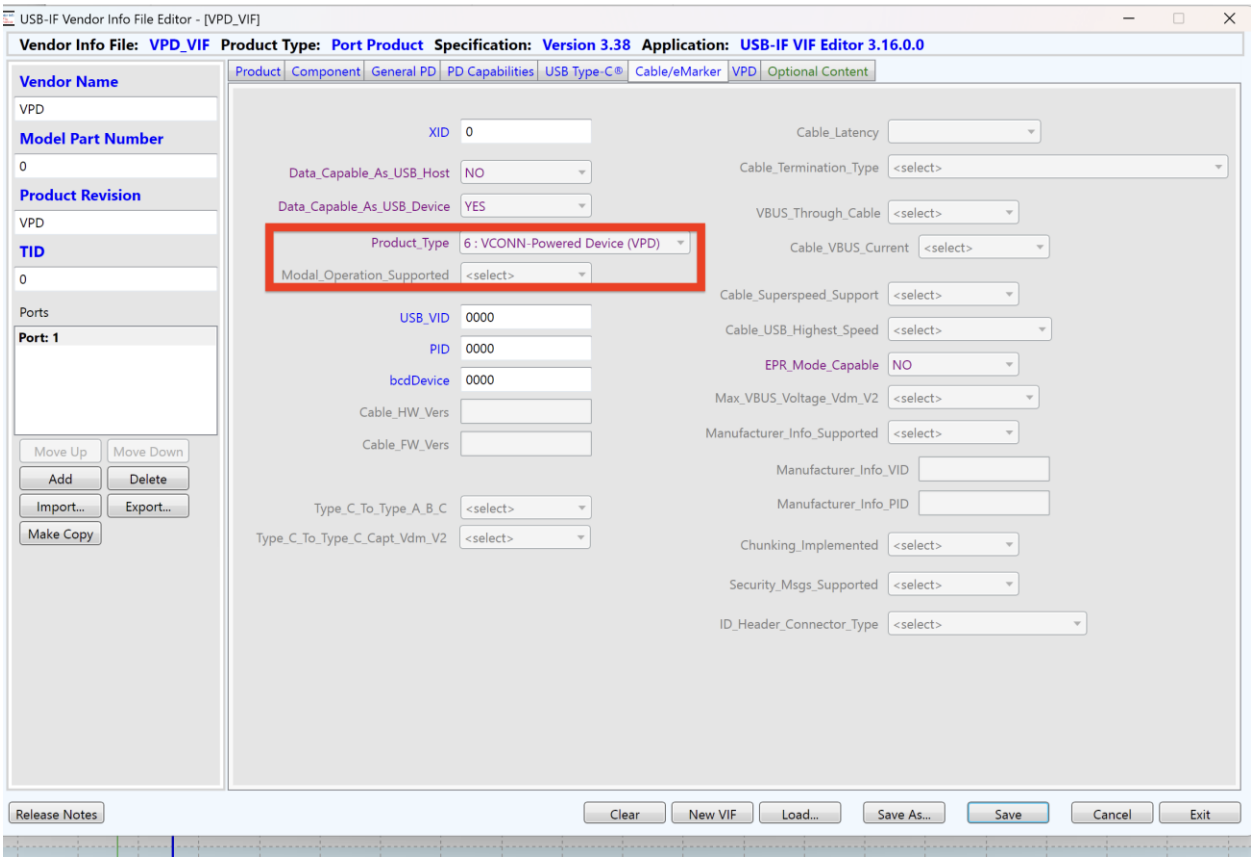
b. VPD_Charge_Through_Support = 0: Reserved 000000b

9. B6…1
    a. VPD_Charge_Through_Support = 1: VPD_Ground_Impedance
    b. VPD_Charge_Through_Support = 0: Reserved 000000b
10. B0 VPD_Charge_Through_Support

4. The Tester sends a *Discover SVIDs* Command to the UUT, using SOP'. The Tester checks the response message: [TEST.PD.VDM.CBL.1#2]
    a. The check fails if the UUT responds anything but a *Discover SVIDs* Response message.
    b. If the UUT responds with "Responder BUSY" message, the Tester resends *Discover SVIDs* Command for *nBusyCount* times every *tVDMBusy* min. Once the Tester has repeated *Discover SVIDs* for *nBusyCount* times, the check fails, and the test ends here.
    c. If the UUT responds with a "Responder NAK" message,
        a. the check passes if Product Type (ID Header B29…27) = VPD and the test stops here.
        b. the check fails if VIF field Modal_Operation_Supported = YES.
    d. If the UUT responds with a "Responder ACK" message, the check fails if VIF field Modal_Operation_Supported = NO.
5. The Tester evaluates the number of SVIDs in the *Discover SVIDs* ACK message and checks the following: [TEST.PD.VDM.CBL.1#3]
    a. Each discovered SVID appears in VIF CableSVIDList. SVID's position in the VDO is checked according to SVID_Fixed value in VIF.
        i. If SVID_Fixed is set to YES, the order of SVIDs in the VDO matches the order of CableSVID elements in VIF CableSVIDList.
        ii. If SVID_Fixed is set to NO, each SVID in the VDO list can be in any position in the VIF's CableSVIDList.
    b. The number of SVIDs discovered lies between Num_SVIDs_min and Num_SVIDs_max.
    c. For each VDO, the Tester checks if B15…0 (SVID n+1) are set to zeros, this marks the last VDO.
    d. For each VDO, the Tester checks if B31…16 (SVID n) are set to zeros, then B15…0 shall also be zeros, and this marks the last VDO.
    e. If the *Discover SVIDs* ACK message has 6 VDOs, and B15…0 in VDO 6 is non-zero, then repeat Step-4 until the Tester finds the last VDO.
6. For each SVID:
    a. The Tester sends a *Discover Modes* Command to the UUT, using SOP'.
    b. The Tester checks the response: [TEST.PD.VDM.CBL.1#4]
        i. The check fails if the UUT responds anything but a *Discover Modes* Response message.
        ii. If the UUT responds with "Responder BUSY" message, the Tester resends *Discover Modes* Command for *nBusyCount* times every *tVDMBusy* min. Once the Tester has repeated *Discover Modes* for *nBusyCount* times, the check fails, and the test ends here.
        iii. If the UUT responds with a "Responder ACK" message, the check fails if VIF field Modal_Operation_Supported = NO.
        iv. The number of Modes discovered lies between SVID_Num_Modes_Min and SVID_Num_Modes_Max.
        v. If the UUT responds with a "Responder NAK" message, the check fails if VIF field Modal_Operation_Supported = YES. The test ends here.
        vi. Check that each Mode discovered matches a CableSVIDMode element in the VIF CableSVIDModeList The procedure for matching them is as follows:
            1. If SVID_Modes_Fixed is set to YES, the match is done by simply comparing the value in the VDO with the corresponding value in the SVID_Mode_Recog_Value. Both lists should be in the same sequence.
            2. If SVID_Modes_Fixed is set to NO, the value in the VDO is bitwise ANDed with each SVID_Mode_Recog_Mask and the result is compared with the corresponding SVID_Mode_Recog_Value in turn until a match is found.
7. For every Mode in each SVID:
    a. The Tester sends *Enter Mode* to the UUT, using SOP'. If the corresponding VIF CableSVIDMode element contains SVID_Mode_VDO, the tester shall append SVID_Mode_VDO to the VDM.
    b. The Tester checks the response: [TEST.PD.VDM.CBL.1#5]

i. If the UUT responds with “Responder BUSY”, the check fails.
ii. If the UUT responds with “Responder NAK” the text continues.
iii. If the UUT responds with “Responder ACK” to *Enter Mode*, the Tester then sends an *Exit Mode* Command to the UUT using SOP’. The check fails if the UUT does not respond with “Responder ACK” to *Exit Mode*.

8. The Tester sends *Attention* to the UUT using SOP’. The check fails if the UUT does not ignore the message. [TEST.PD.VDM.CBL.1#6]


USB-IF Vendor Info File Editor - [VPD_VIF]
Vendor Info File: VPD_VIF Product Type: Port Product Specification: Version 3.38 Application: USB-IF VIF Editor 3.16.0.0
Vendor Name
VPD
Model Part Number
0
Product Revision
VPD
TID
0
Ports
Port: 1
Move Up
Move Down
Add
Delete
Import...
Export...
Make Copy
Product
Component
General PD
PD Capabilities
USB Type-C®
Cable/eMarker
VPD
Optional Content
XID 0
Data_Capable_As_USB_Host NO
Data_Capable_As_USB_Device YES
Product_Type 6 : VCONN-Powered Device (VPD)
Modal_Operation_Supported <select>
USB_VID 0000
PID 0000
bcdDevice 0000
Cable_HW_Vers
Cable_FW_Vers
Type_C_To_Type_A_B_C <select>
Type_C_To_Type_C_Capt_Vdm_V2 <select>
Cable_Latency
Cable_Termination_Type <select>
VBUS_Through_Cable <select>
Cable_VBUS_Current <select>
Cable_Superspeed_Support <select>
Cable_USB_Highest_Speed <select>
EPR_Mode_Capable NO
Max_VBUS_Voltage_Vdm_V2 <select>
Manufacturer_Info_Supported <select>
Manufacturer_Info_VID
Manufacturer_Info_PID
Chunking_Implemented <select>
Security_Msgs_Supported <select>
ID_Header_Connector_Type <select>
Release Notes
Clear
New VIF
Load...
Save As...
Save
Cancel
Exit